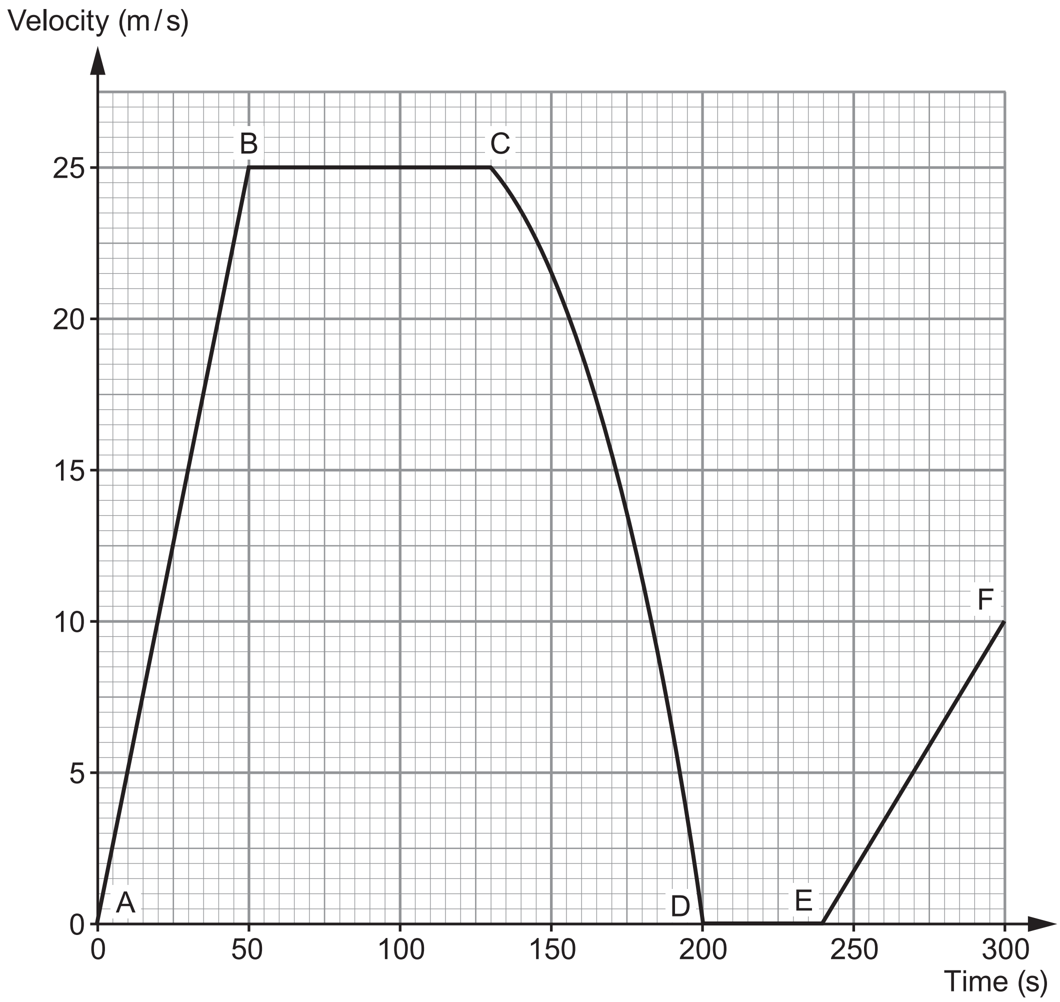
7. The velocity-time graph below shows part of the motion of an empty school bus.



(a) Use values from the graph to describe the motion of the bus over the time shown. [Note that no calculations are required as part of your answer.] [6 QER]

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Examiner only

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(b) Refer to the graph to answer the following questions. The mass of the bus is 10 000 kg.

(i) Use an equation from page 2 to calculate its change in momentum between A and B. [2]

Change in momentum = ... kg m/s

(ii) Use an equation from page 2 to calculate the resultant force that acts on the bus between A and B. [2]

Resultant force = ... N

(iii) Use the equation:

distance = speed × time

to calculate the distance travelled between B and C.

[2]

Distance = ... m

Examiner only

(iv) Another identical bus travels the same distance as in part (iii). However, it travels at a slower, constant speed. State **two** ways that **its line on the graph** would be different from the one shown. [2]

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(c) Explain, using the idea of inertia, how the acceleration of a bus filled with school children would compare with the acceleration of the empty bus. [2]

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